

Program 4:

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <string.h>

int main() {
    int fd[2];
    pid_t pid;

    char message[] = "Hello from Parent";
    char buffer[100];

    pipe(fd);

    pid = fork();

    if (pid == 0) {
        close(fd[1]);

        read(fd[0], buffer, sizeof(buffer));

        printf("Child received: %s\n", buffer);

        close(fd[0]);
    }
    else {
        close(fd[0]);

        write(fd[1], message, strlen(message) + 1);

        close(fd[1]);

        wait(NULL);
    }

    return 0;
}
```

Output:

```
crazyguy64@CrazyGuy64:~/practice$ nano program4.c
crazyguy64@CrazyGuy64:~/practice$ gcc program4.c -o program4
crazyguy64@CrazyGuy64:~/practice$ ./program4
Child received: Hello from Parent
```

Fifo_write.c:

```
#include<stdio.h>
#include<fcntl.h>
#include<unistd.h>
#include<string.h>

int main(){
int fd;
char message[] = "Hello from Writer";

fd = open("myfifo", O_WRONLY);

write(fd, message, strlen(message) + 1);

close(fd);

printf("Message sent\n");

return 0;
}
```

Output:

```
crazyguy64@CrazyGuy64:~/practice$ nano write.c
crazyguy64@CrazyGuy64:~/practice$ gcc write.c -o write
crazyguy64@CrazyGuy64:~/practice$ ./write
Message sent
```

Fifo_Create:

```
#include<stdio.h>
#include<sys/types.h>
#include<sys/stat.h>

int main(){
if(mkfifo("myfifo", 0666) == -1){
perror("mkfifo");
return 0;
}
printf("FIFO creates\n");
return 0;
}
```

Output:

```
crazyguy64@CrazyGuy64:~/practice$ nano fifo.create.c
crazyguy64@CrazyGuy64:~/practice$ gcc fifo.create.c -o fifo.create
crazyguy64@CrazyGuy64:~/practice$ ./fifo.create
FIFO creates
```

Fifo_write:

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <string.h>

int main(){
    int fd;
    char message[] = "Hello form Writer";
    fd = open("myfifo", O_WRONLY);
    write(fd,message, strlen(message)+1);
    close(fd);
    printf("Message sent\n");
}
```

Output:

```
crazyguy64@CrazyGuy64:~$ cd practice
crazyguy64@CrazyGuy64:~/practice$ nano fifo.write.c
crazyguy64@CrazyGuy64:~/practice$ gcc fifo.write.c -o fifo.write
crazyguy64@CrazyGuy64:~/practice$ ./fifo.write
hello
hi
namaste
|
```